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**European Patent Application 16802532.8**  
**USE OF PTERIDINONE DERIVATIVE SERVING AS EGFR**  
**INHIBITOR**  
**East China University of Science and Technology**

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Responsive to the communication dated 11 November 2020:

An amended set of 8 claims is submitted in mark-up and clean copy which should be the basis for the further prosecution.

**Amendments**

Three compounds of former claim 3 have been incorporated into new claim 1 as alternatives to the compounds of general formula I. This amendment is based for example on PCT claim 3 for the first two compounds and on page 30 of the English translation of the PCT application for compound 12.

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Former claims 3 and 5 have been made dependent on new claim 1. These amendments are based at least on the particular embodiment starting on page 14, line 23 and the preferred embodiment starting on page 18, line 9 of the English translation of the PCT application, respectively.

### **Clarity**

The clarity objection regarding claims 3 and 5 is overcome by the amendment indicated above.

### **Unity**

Amended claim 1 fulfils the requirements of unity, because all compounds claimed in claim 1 are alternatives having common activity and a common structure. As already indicated in the response dated 6 August 2020, the compounds of general formula I and compounds 4, 5 and 12 (which are the compounds now introduced into claim 1) show excellent EGFR inhibitory activity.

### **Novelty**

It was objected to in the office action that the compounds of former claims 3 and 5 with  $R^1$  being hydrogen (compounds 4, 5 and 12 which are now part of claim 1) are not novel over D4. It is not agreed. While D4 in general may disclose  $R^1$  being hydrogen, the specific compounds now incorporated into claim 1 are not directly and unambiguously disclosed in D4. It is noted that e.g. the methoxy group of compound 4 of the present application is in a different position than the methoxy group of compound 32 of D4.

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The three compounds newly introduced into claim 1 are not anticipated by D4 and the subject matter of claim 1 is novel.

**Inventive step**

As already indicated in the previous response dated 6 August 2020, the compounds 4, 5 and 12 show excellent EGFR inhibitory activity. For example, the IC<sub>50</sub> values of inhibitory activity on proliferation on cells (H1975) of compounds 4, 5 and 12 are 0.009, 0.006, and 0.006  $\mu$ M, respectively. This was completely unpredictable when starting from the compounds of D4. For example, compound 19 of D4 shows an IC<sub>50</sub> value of inhibitory activity on the proliferation of H1975 of only 8.87  $\mu$ M.

Therefore, providing the compounds newly introduced into claim 1 was not obvious for the person skilled in the art and the subject matter of new claim 1 is based on an inventive step.

It is kindly requested to indicate the patentability of the present set of claims. An adapted description will subsequently be submitted.

As a precaution, just in case the Examining Division intends to refuse the application, oral proceedings are requested.

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Enclosures: amended claims (mark-up)  
amended claims (clean copy)